

Standard Operating Procedure: Centrifugal Pump Startup and Shutdown

SOP ID: SOP-PUMP-07 | Rev: 3 | Approved: 2024-03-01 | Applies to: P-101, P-102, P-201

1. Pre-Startup Checks

- 1.1 Confirm maintenance work order is closed and equipment returned to service.
- 1.2 Verify all isolation valves on suction line are fully open.
- 1.3 Check mechanical seal flush line is open; flush pressure must read 0.8 to 1.2 bar.
- 1.4 Confirm bearing housing oil level is between MIN and MAX marks on sight glass.
- 1.5 Check coupling guard is fitted and all bolts are torqued.
- 1.6 Set discharge valve to 25 percent open. Never start against a fully closed or fully open discharge valve.
- 1.7 Manually rotate pump shaft to confirm free rotation. Resistance indicates seized bearing or locked seal - do not attempt to start.

2. Startup Sequence

- 2.1 Energise motor from MCC panel. Confirm green running indication.
- 2.2 Slowly open discharge valve to duty point over 30 seconds.
- 2.3 Confirm discharge pressure stabilises at design value within plus or minus 10 percent.
- 2.4 Measure vibration at bearing housing within 5 minutes. Must be below 4.5 mm/s per OISD-137. Shut down immediately if reading exceeds 4.5 mm/s.
- 2.5 Check bearing temperature within 15 minutes. Must be below 70 deg C.
- 2.6 Inspect gland area for leakage. Maximum acceptable: 10 drops per minute.
- 2.7 Record startup in maintenance register with date, time, and technician signature.

3. Normal Shutdown

- 3.1 Partially close discharge valve to 25 percent open.
- 3.2 De-energise motor from MCC panel.
- 3.3 Close discharge valve fully once pump comes to a complete stop.
- 3.4 Close suction valve if pump is to be isolated for maintenance.
- 3.5 Depressurise and drain casing before any mechanical work begins.

4. Emergency Shutdown

- 4.1 De-energise motor immediately from nearest isolation point.
- 4.2 Do not attempt to restart until root cause of emergency is identified.
- 4.3 Raise maintenance work order in CMMS system within 30 minutes.
- 4.4 Notify shift supervisor immediately.

5. Safety Precautions

- 5.1 Follow LOTO procedure HSE-LOTO-001 before any mechanical work.
- 5.2 Mandatory PPE: safety glasses, chemical-resistant gloves, steel-toed boots.
- 5.3 Never open pump casing while pressurised. Depressurise to zero before opening.
- 5.4 Refer to MSDS for the product being pumped before any maintenance task.